Lecture 2:
Introduction to ERP systems

Business Applications (Betriebswirtschaftliche Anwendungen)

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Course content

1	Course Kickoff
2	Introduction to ERP systems
3	SAP ERP
4	Principles of ERP systems
5	Modern ERP concepts
6	Fundamentals of AI agents
7	AI agent ecosystem & Retro
8	AI agents in business workflows
9	Agentic ERP in S/4 HANA
10	OpenClaw & Test preparation
11	E-Test

Learning objectives

At the end of the lecture, the students are able to …

- explain the **evolution of ERP systems** from MRP to modern, cloud-based platforms
- describe **key characteristics and development principles** of modern ERP systems
- outline the **business value of ERP** along end-to-end processes
- analyze the **global ERP market**, including vendor shares, regional differences and cloud vs. on-premise dynamics
- compare the **strategies of leading ERP vendors**
- assess **future ERP trends**

Agenda

1. **Challenges and Characteristics of ERP Systems**
2. ERP Market Analysis
3. ERP Future Trends

ERP evolution

MRP

1960s

Material Requirements Planning

Narrow manufacturing focus

MRP II

1983

Manufacturing Resource Planning

Cross-functional coordination

ERP

1990s

Enterprise Resource Planning

Cross-industry, client-server architecture (SAP R/3)

Modern ERP

2010+

ERP systems with modern technology

Cloud, AI, predictive analytics, real-time anywhere access, etc.

From inventory tracking to intelligent, cloud-based enterprise platforms

Source: Sarferaz (2022)

Characteristics of modern ERP systems

Performance & Scalability
In-memory database, one-system OLTP + OLAP

User Experience
Intuitive, any device, any time (e.g., SAP Fiori UX)

Extensible Architecture
In-app and side-by-side, extensibility via APIs

Standardized Implementation
Pre-configured packages, elf-service tools

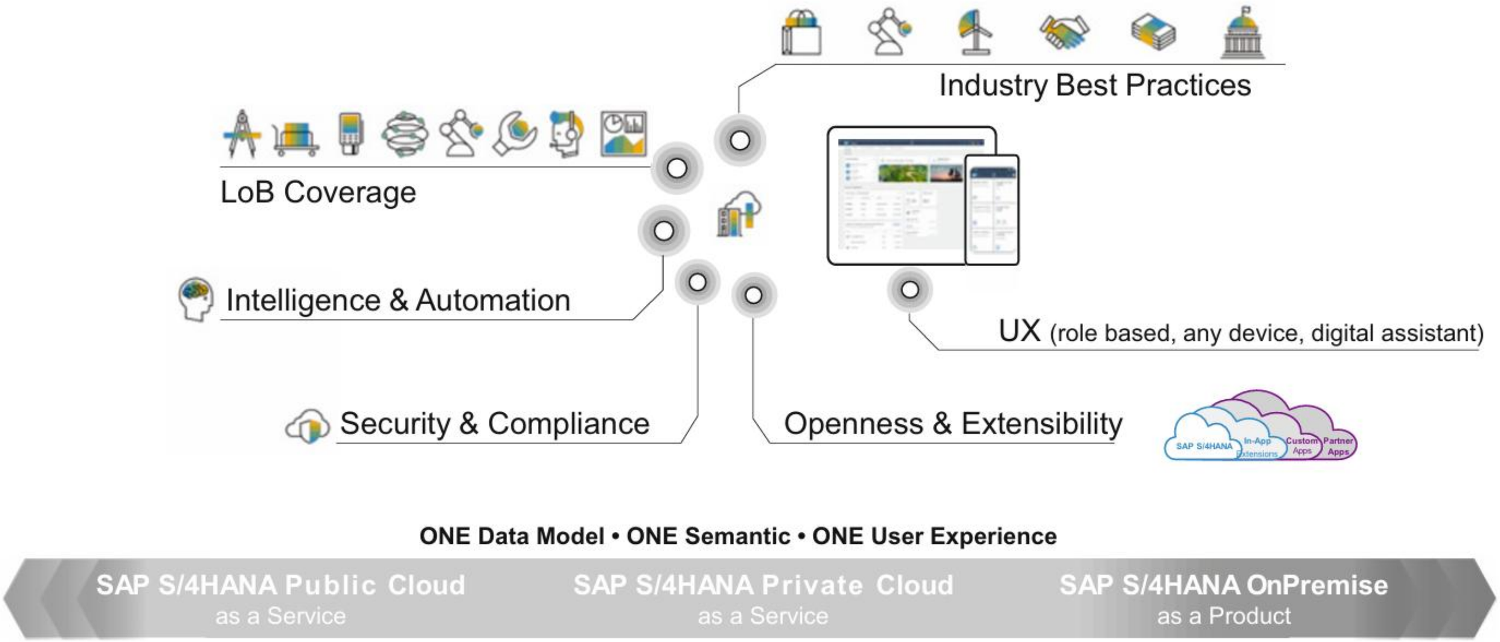
Intelligent ERP Processes
AI, ML and predictive analytics in processes

Cloud & On-Premise
SaaS, public/private cloud, hybrid options

Security & Compliance
Data protection, data isolation, compliance

Source: Sarferaz (2022)

SAP S/4 HANA key capabilities



LoB: Lines of Business

Source: Sarferaz (2022)

SAP S/4 HANA development principles

1 Stable but Flexible Core
Clean core for faster deployment; in-app and side-by-side extensibility via stable APIs

2 Principle of "ONE"
Deprecation of redundant data models, frameworks, and UIs for simplification – instead ONE data model, ONE semantic, ONE UX

3 Service Orientation
Virtual Data Model (VDM) for cross-references; RESTful services for SAP Fiori apps and open standards

4 Modularization
Container technology and logical data model across services; cloud and on-premise hybrid support

5 Cloud First, but not Only
Pure SaaS with multitenancy, zero downtime, fully dynamic tenant density

6 Semantic Compatibility
Homogeneous data models across on-premise and cloud, smooth migration path from SAP ERP

Source: Sarferaz (2022)

Business value of ERP

Lead to Cash
Customer intent to revenue collection

Source to Pay
Sourcing goods to invoice payment

Design to Operate
Product design to maintenance

Recruit to Retire
Workforce planning to retirement

Strategic Value

- Industry standardization of business processes reduces total cost of ownership (TCO)
- Faster innovation adoption
- Cloud-based digital transformation

Operational Value

- Real-time insights and embedded analytics
- Predictive planning and simulation
- Process automation (RPA) frees users from manual routine tasks

Source: Sarferaz (2022)

Agenda

1. Challenges and Characteristics of ERP Systems
2. **ERP Market Analysis**
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Activity: Who are the typical players in the ERP market?

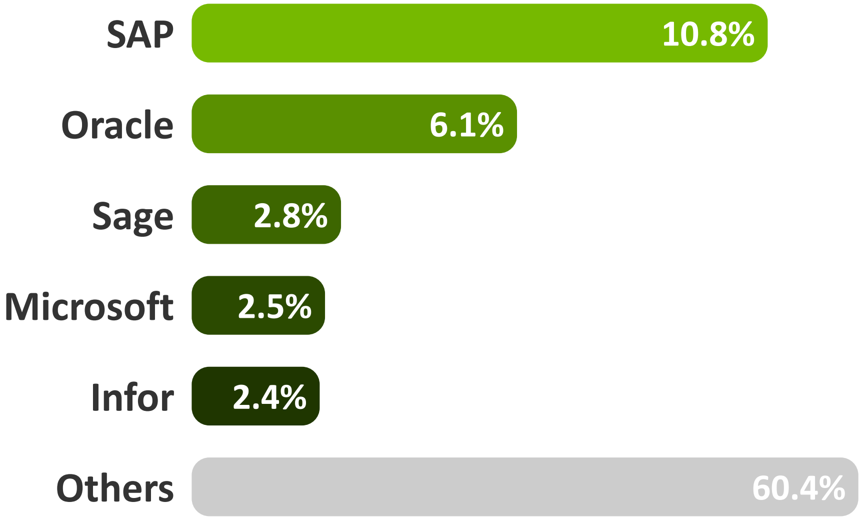
- Own brainstorming without researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)

ERP market volume (2019)

€53B

Worldwide ERP Market

Source: Gartner/Pang et al. 2020



Key insight: Top 10 vendors account for ~30% of the market—60%+ held by smaller vendors

Source: Sarferaz (2022)

Regional ERP market shares (2019)

North America €23.5B

1. Oracle 8.7%
2. **SAP** 7.3%
3. Infor 3.1%

EMEA €18B

1. **SAP** 15.1%
2. Sage 4.8%
3. DATEV 4.6%

APJ & China €9.4B

1. **SAP** 10.1%
2. Yonyou 4.5%
3. Oracle 4.2%

Latin America €2B

1. **SAP** 17.5%
2. Totvs 13.2%
3. Oracle 6.5%

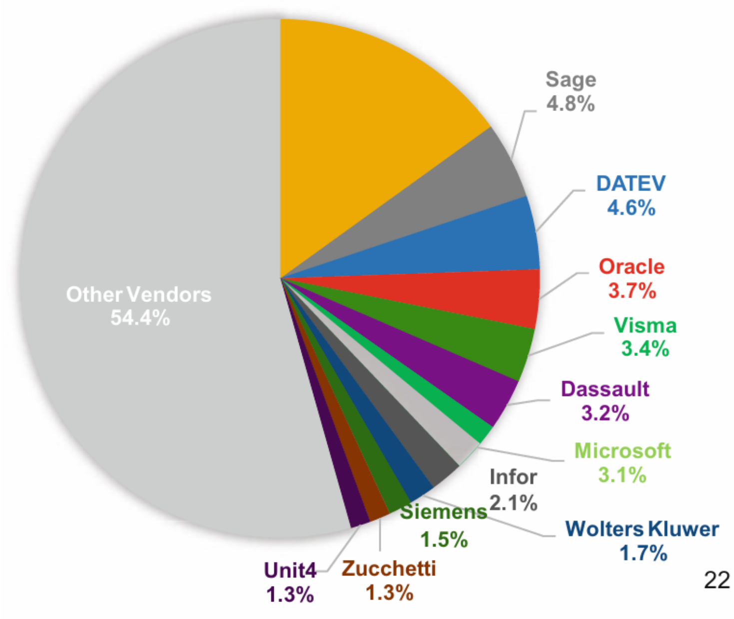
Key insight: SAP leads in 3 of 4 regions; Oracle leads in North America (8.7% vs SAP 7.3%)

Source: Sarferaz (2022)

Deep-Dive: Regional ERP market share (2019) (1/2)

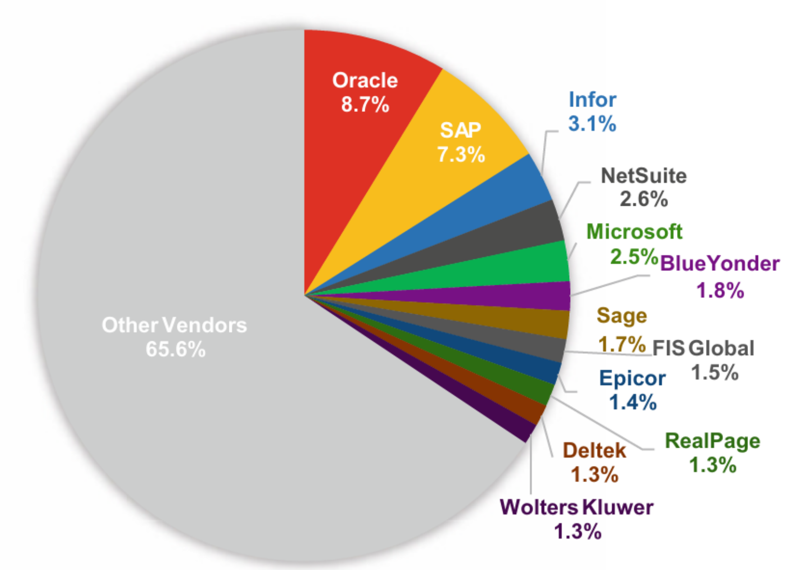
EUROPE MIDDLE EAST AFRICA

MARKET SIZE: €18B



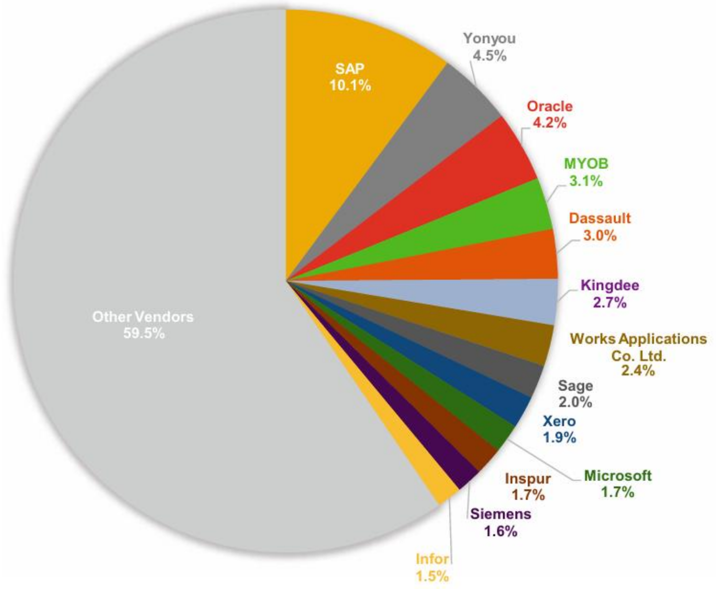
NORTH AMERICA

MARKET SIZE: €23.5B



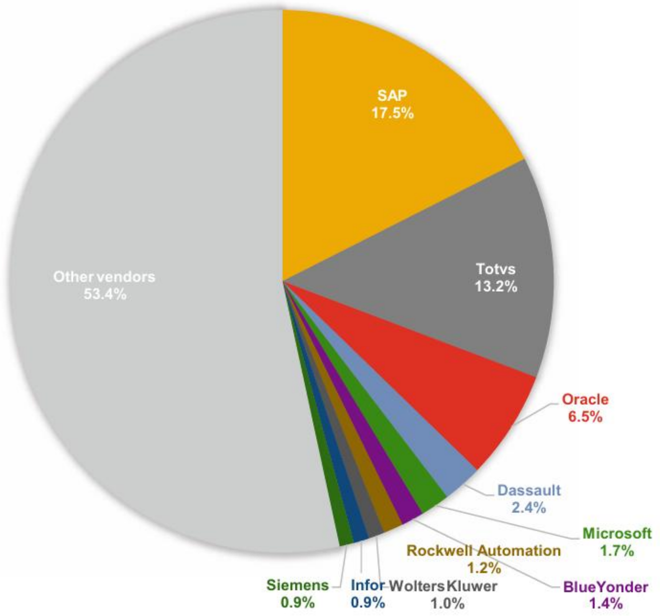
Source: Sarferaz (2022)

Deep-Dive: Regional ERP market share (2019) (2/2)



MARKET SIZE: €9.4B

ASIA PACIFIC JAPAN CHINA



MARKET SIZE: €2.0B

LATIN AMERICA

Source: Sarferaz (2022)

ERP competitors and strategies

S
SAP
- Market leader since 1972
- Migrate to S/4HANA Cloud
- Customer experience via Qualtrics

O
Oracle
- Cloud infrastructure
- NetSuite acquisition (2016)
- Analytics & AI

W
Workday
- Cloud-first finance
- Analytics and predictions
- ~50% YoY growth

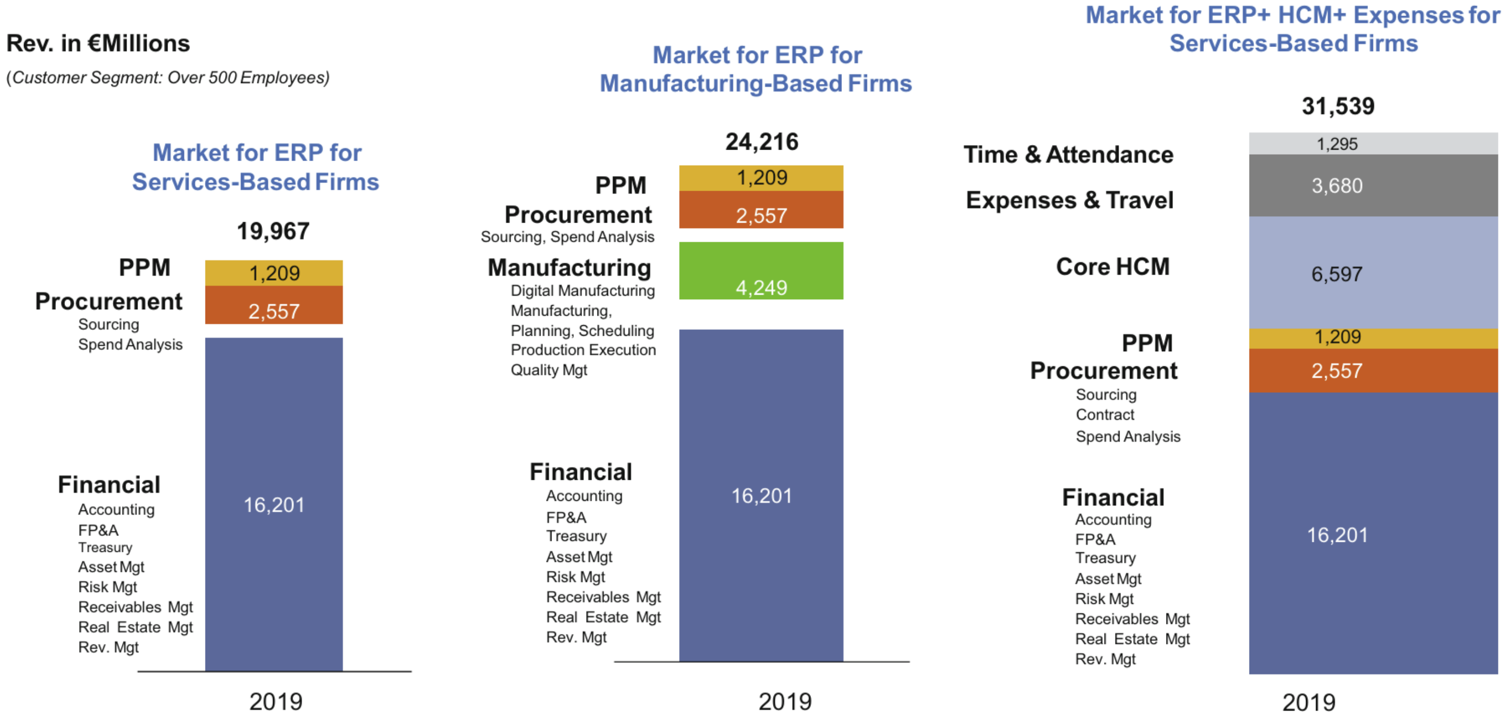
M
Microsoft
- Azure bundling
- Dynamic products
- AI platform
- Retail focus

I
Infor
- Upgrade to cloud since 2010
- Industry-specific (auto, fashion, F&B)
- Rapid dev

Key insight: All major vendors converge on cloud-first, AI and customer-centric service models

Source: Sarferaz (2022)

ERP market size per type of customer use case (2019)



Source: Sarferaz (2022)

ERP market forecast

Cloud ERP

- €15.8B (2019) → €33.9B (2024 projected)
- YoY growth: 25.9% (2019) → 12.2% (2020) due to Covid → 22.1% (2024)
- Overtook on-premise revenue in 2021
- Driven by, e.g., scalability, reduced costs

On-Premise ERP

- Declining revenue year-over-year
- YoY growth: −9.9% (2020), −6% (2021)
- Projected: +0.2% by 2024 (stabilizing)
- Relevant for specific use cases

Key growth drivers

AI & Big Data | IoT Integration | Mobile Cloud Apps | SMB Adoption

Source: Sarferaz (2022)

Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes

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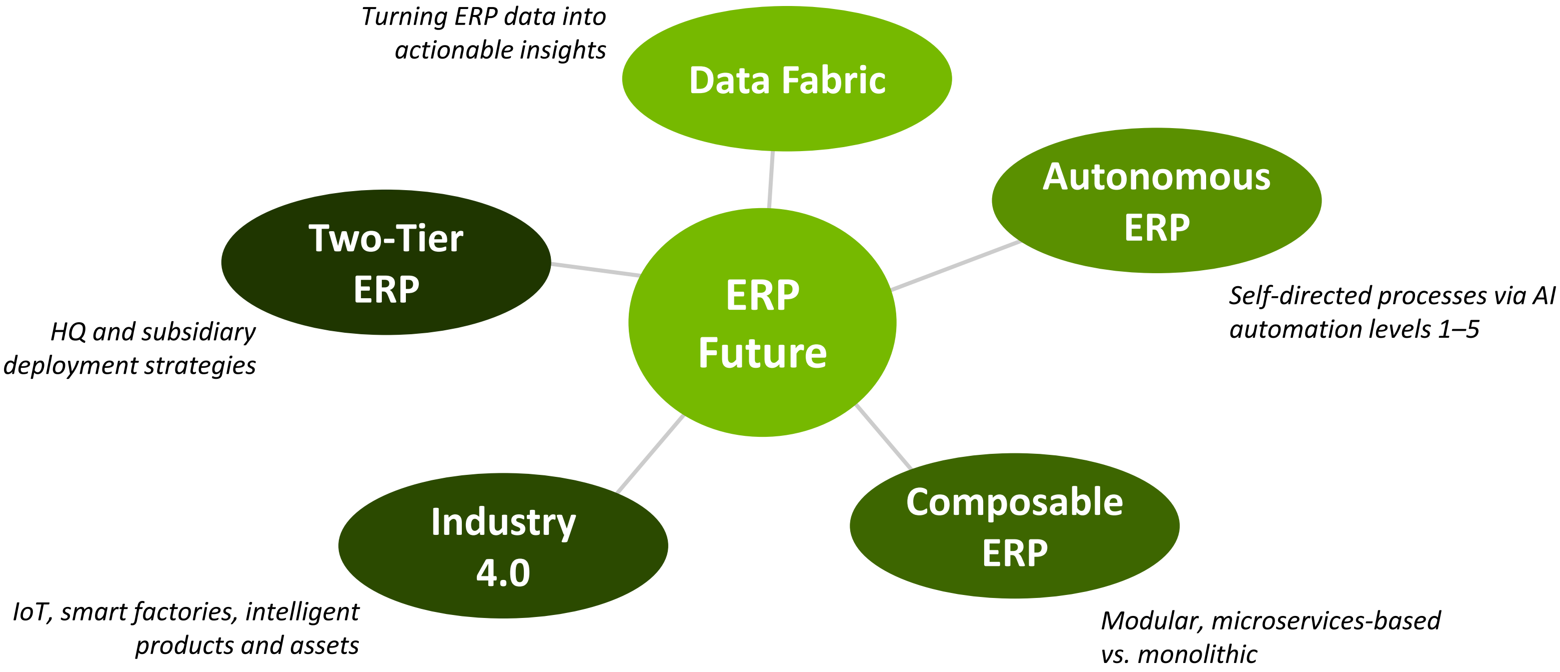
Trends impacting the ERP market



Source: Sarferaz (2022)

Data fabric and data science in ERP

Problem Definition → Data Acquisition → Data Preparation → Data Modeling → Insight Reporting

ERP data applications

Sales and Finance

Time series analysis, revenue forecasting, financial event prediction

IoT and Maintenance

Predictive maintenance from sensor data, automated anomaly detection

Process Automation

NLP for invoice extraction, computer vision for quality checks, enterprise intelligence

Key insight: ~40 zettabytes in 2020 (300x growth since 2012). Orgs using data science in ERP gain competitive advantage through deeper customer insights and process optimization.

Source: Sarferaz (2022)

Autonomous ERP: Four dimensions of process automation

Vision: Self-directed business processes and self-diagnostic operations

Data Acquisition	Information Analysis	Decision Making	Action Execution
L5 AI-based extraction	L5 Cognitive (self-learning)	L5 Autonomous	L5 Autonomous
L4 Conversational AI	L4 Prescriptive	L4 Recommending	L4 Recommending
L3 Integration + exceptions	L3 Predictive	L3 System-informed	L3 System-guided
L2 Manual + integration	L2 Diagnostic	L2 Event-based	L2 Event-triggered
L1 Manual entry	L1 Descriptive	L1 Manual	L1 Manual

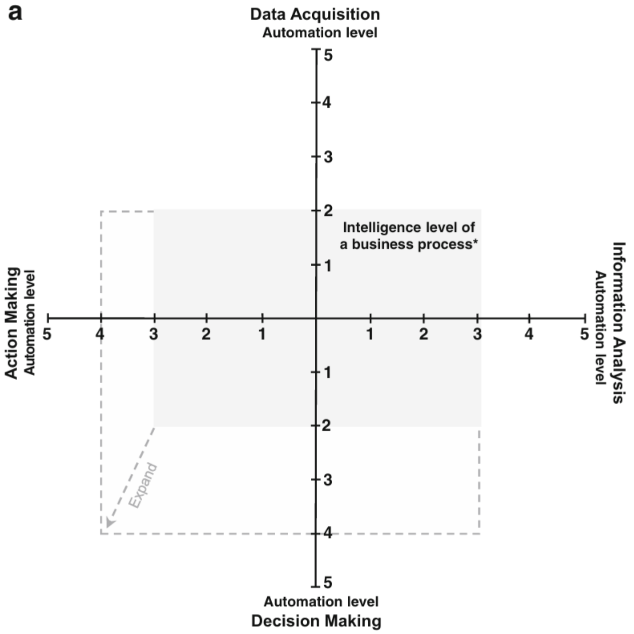
Autonomy

Source: Sarferaz (2022)

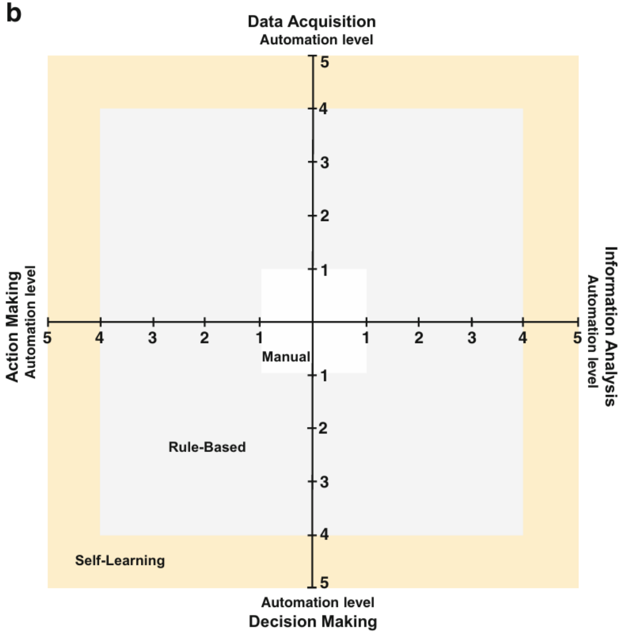
Activity: Name examples of applications with high autonomy.

- Own brainstorming with researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)

Autonomous ERP: Framework



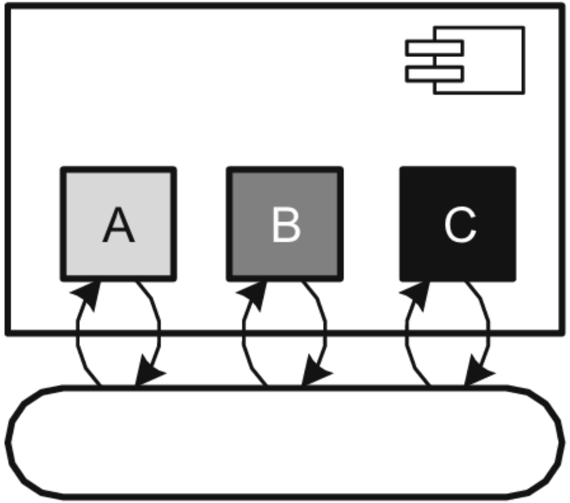
a) Four dimensions of process automation



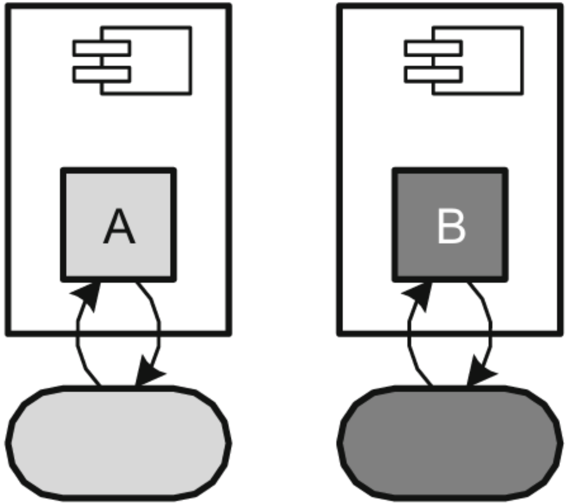
b) Categorization of techniques based on automation level

Source: Sarferaz (2022)

Composable ERP: Architecture approaches (1/2)



Monolithic application



Microservice-based application

Source: Sarferaz (2022)

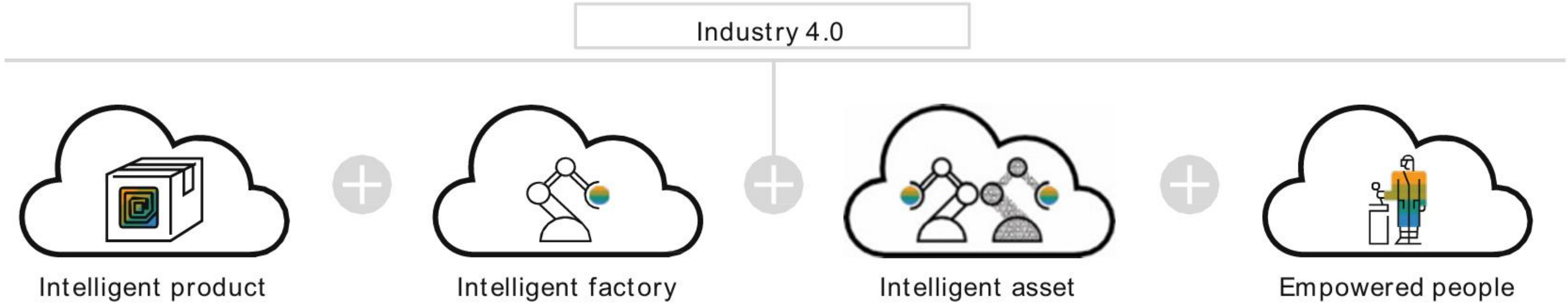
Composable ERP: Architecture approaches (2/2)

Monolithic (tightly coupled)	**Microservices** (loosely coupled)
✓ **Strengths**	✓ **Strengths**
• Shared DB schema = strong consistency	• Independent deployment
• Single transaction scope	• High scalability per service
• Simple extensibility	• Technology flexibility
• Good for ERP core processes	• Good for specific workloads
✗ **Limitations**	✗ **Limitations**
• Full system upgrades required	• Eventual consistency only
• Less granular scaling	• Complex cross-service analytics

SAP S/4HANA approach: Complementary—monolithic core for tightly coupled ERP processes and microservices (Docker/Kubernetes) for scalable deep-learning use cases

Source: Sarferaz (2022)

Industry 4.0 and IoT in ERP



Intelligent product
- Designed and configured to precisely meet customer needs
- Mass customization

Intelligent factory
- Autonomous operations
- Data-driven processes
- Customized products at scale

Intelligent asset
- Dynamically maintained
- Predictive maintenance via sensor data integration

Empowered people
- AR/VR guidance
- Tools and information for best work
- Safety management

IoT-ERP integration: IoT endpoints: 3.96B (2018) → 4.81B (2019), ~20%+ annual growth; ERP as data hub and digital shadow of value chain; strategic prio for 68% of manufacturers

Source: Sarferaz (2022)

Two-tier ERP strategy: Overview

Tier 1 — Headquarters
SAP S/4HANA On-Premise or Private Cloud

Subsidiary A
e.g., Local Finance,
Local Manufacturing

Subsidiary B
e.g., Local Finance,
Warehouse Management

Subsidiary C
e.g., Local Finance,
Retail Operations

Tier 2
S/4HANA Cloud

Public Cloud
Shared resources, SaaS model

Private Cloud
Dedicated resources, single-tenant

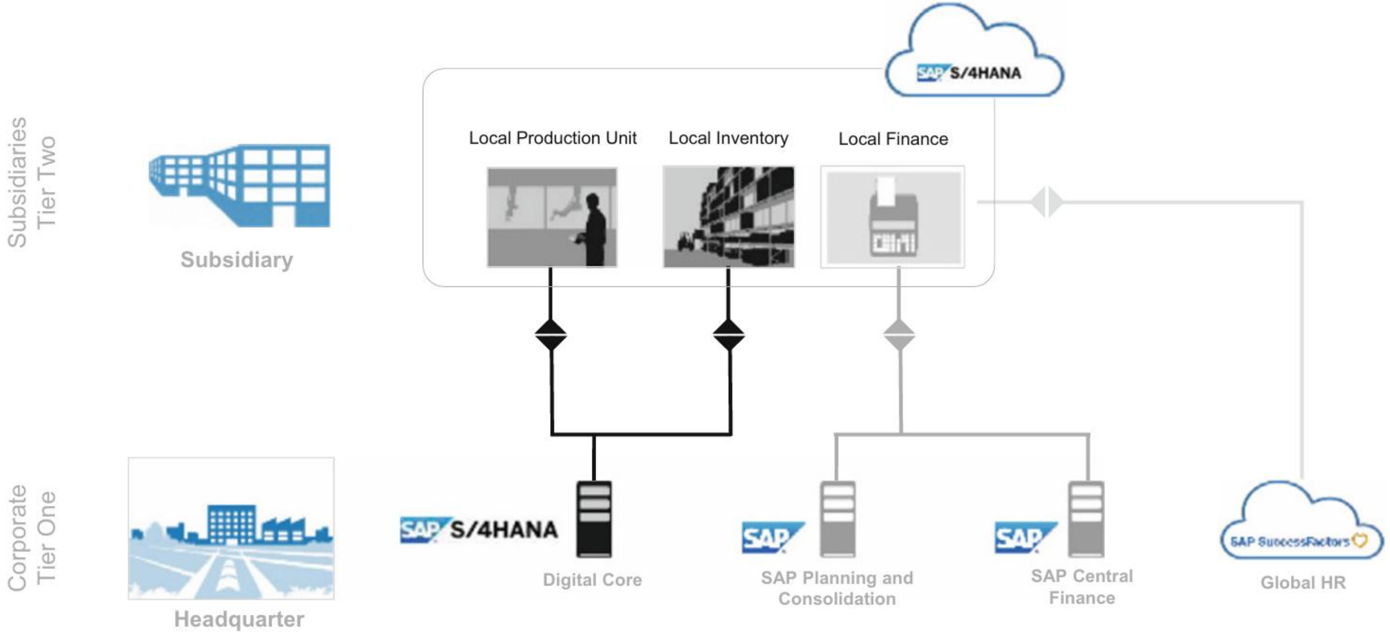
On-Premise
Company server, full control

Key benefit of single-vendor solution: Same code line, shared data structures and APIs—template-based subsidiary rollouts with semantic compatibility

Source: Sarferaz (2022)

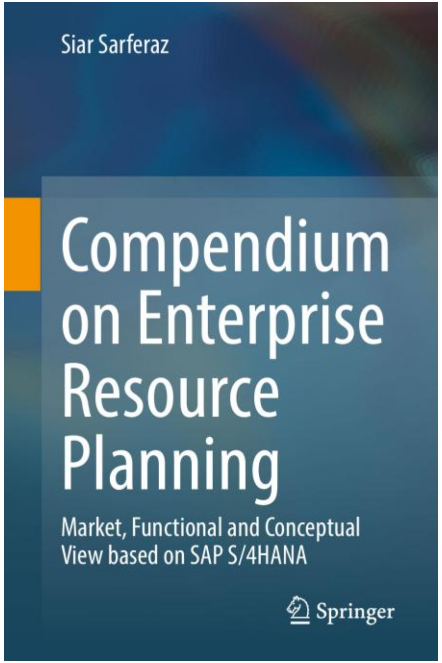
Two-tier ERP strategy: SAP example

SAP S/4HANA Cloud and on-premise for headquarter-subsidiary scenario



Source: Sarferaz (2022)

Reference



Sarferaz, S. (2022): Compendium on Enterprise Resource Planning: Market, Functional and Conceptual View Based on SAP S/4HANA. Springer Nature Switzerland AG.

Thank you for your attention

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